CIC truth seeker dataset 2023 (TruthSeeker2023)

This project aims to create the largest ground truth fake news analysis dataset for real and fake news content in relation to social media posts. Below illustrates the major contributions of the TruthSeeker dataset to the current fake news dataset landscape:

One of the most extensive benchmark datasets with more than 180,000 labelled Tweets.

- Three-factor active learning verification method which involved utilising 456 unique, highly skilled, Amazon Mechanical Turkers for labelling each Tweet.
- To understand patterns and characteristics of Twitter users, three auxiliary social media scores are also introduced: Bot, credibility, and influence score.
- Conducted comprehensive analyses and evaluations on the TruthSeeker dataset, including the establishment of deep learning-based detection models, clustering-based event detection, and exploration of the relationship between tweet labels and the characteristics of online creators/spreaders.

- The application of multiple BERT-based models to assess the accuracy of real/fake tweet detection.

Configuration chart (../_assets/images/truth-seeker-configuration-2023.jpg)

The data for the Truth Seeker and Basic ML dataset were generated through the crawling of tweets related to Real and Fake news from the Politifact Dataset. Taking these ground truth values and crawling for tweets related to these topics (by manually generating keywords associated with the news in question to input into the twitter API), we were able to extract over 186,000 (before final processing) tweets related to 700 real and 700 fake pieces of news.

Taking this raw tweet data, we then used crowdsourcing in the form of Amazon Mechanical Turk to generate a majority answer to how closely the tweet agrees with the Real/Fake news source statement. After, a majority agreement algorithm is employed to designate a validity to the associated tweets in both a 3 and 5 category classification column.

This results in one of the largest ground truth datasets for fake news detection on twitter ever created. The **TruthSeeker Dataset**. Then we also generated a **dataset of features** from the tweet itself and the metadata of the user who posted the related tweet. Allowing the user to have the option to use both deep learning models as well as classical machine learning techniques.

Feature dataset

Tweet Feature Category: Text Features

Name	Description

unique count	number of unique, complex words
total count	total number of words
ORG percent	Percent of text including spaCy ORG tags
NORP percent	Percent of text including spaCy NORP tags
GPE percent	Percent of text including spaCy GPE tags
PERSON percent	Percent of text including spaCy PERSON tags
MONEY percent	Percent of text including spaCy MONEY tags
DATA percent	Percent of text including spaCy DATA tags
CARDINAL percent	Percent of text including spaCy CARDINAL tags
PERCENT percent	Percent of text including spaCy PERCENT tags
ORDINAL percent	Percent of text including spaCy FAC tags
LAW percent	Percent of text including spaCy LAW tags
PRODUCT percent	Percent of text including spaCy PRODUCT tags
EVENT percent	Percent of text including spaCy EVENT tags
TIME percent	Percent of text including spaCy TIME tags
LOC percent	Percent of text including spaCy LOC tags
ORG percent	Percent of text including spaCy ORG tags
WORK OF ART percent	Percent of text including spaCy WOA tags
QUANTITY percent	Percent of text including spaCy QUANTITY tags
LANGUAGE percent	Percent of text including spaCy LANGUAGE tags
Max Word	length of the longest word in the sentence
Min Word	length of the shortest word in the sentence
Avg Word Length	average length of words in the sentence

Tweet Feature Category: Lexical Features

Name	Description
present verb	number of present tense verbs
past verb	number of past tense verbs
adjectives	number of adjectives
pronouns	number of pronouns
TO's	number of to usages
determiners	number of determiners
conjunctions	number of conjunctions
dots	number of (.) used
exclamations	number of (!) used
question	number of (?) used
ampersand	number of (&) used
capitals	Number of capitalized letters
quotes	number of quotation makes used
digits	number of digits (0-9) used
long word freq	number of long words
short word freq	number of short words

Tweet Feature Category: Meta-Data Features

Name	Description
followers count	number of followers
friends count	number of friends
favourites count	number of favourites across all tweets
statuses count	number of tweets
listed count	number of tweets the user has in lists
mentions	number of times the user was mentioned
quotes	number of times the user has been quote tweeted
replies	number of replies the user has
retweets	number of retweets the user has
favourites	number of favourites the user has
hashtags	number of hashtags the user has used
URLs	whether the user has a provided a url in relation to their profile
BotScoreBinary	Binary score whether the user is considered a bot or not
cred	credibility score
normalized influence	influence score the user has, normalized

The feature only dataset contains textual and lexical information related to each tweet. As well as metadata information about the user of said tweet. All in all providing over 50 features for training on any classical machine learning model, rather than more advanced deep learning algorithms.

Truth Seeker dataset

3 label majority answers (../_assets/images/truth-seeker-3-label.jpg)

5 label majority answers (../_assets/images/truth-seeker-5-label.jpg)

The TruthSeeker dataset contains the aforementioned tweets, the source statement of news in which the keywords used to source the tweets were created from, the manual keywords of the source statement, the 5 label majority answer of the truthfulness value, and the 3 label majority answer.

This dataset provides the opportunity for training deep learning BERT based models on a large corpus of crowdsourced ground truth tweets. Allowing for fine grain 5 label classification (4 if unknowns are removed) or more general 3 label classification (binary if unknowns are removed). The associated paper shows the possible conversion table that can be used to then assign truthfulness values to the individual tweets.

Directory

The main dataset directory (TruthSeeker2023) contains two separate .csv files:

1. **Truth_Seeker_Model_Dataset:** This file contains the features mentioned in the above TruthSeeker Dataset section. Designed for use with BERT based NLP models.
2. **Truth_Seeker_Model_Dataset_With_TimeStamps:** This file contains the features mentioned in the above TruthSeeker Dataset section. Designed for use with BERT based NLP models with the addition of timestamps for each Tweet.

3. **Features_For_Traditional_ML_Techniques:** This file contains the 50+ features mentioned in the above Feature Dataset section. Designed to be used with classical machine learning techniques which take many features as input rather than generate features from data.
4. **Readme.txt:** This readme file contains descriptions of each feature in both datasets.

Description of features in both datasets

Features_For_Traditional_ML_Techniques

Name	Description
unique_count	number of unique, complex words
total_count	total number of words
ORG_percent	Percent of text including spaCy ORG tags
NORP_percent	Percent of text including spaCy NORP tags
GPE_percent	Percent of text including spaCy GPE tags
PERSON_percent	Percent of text including spaCy PERSON tags
MONEY_percent	Percent of text including spaCy MONEY tags
DATA_percent	Percent of text including spaCy DATA tags
CARDINAL_percent	Percent of text including spaCy CARDINAL tags
PERCENT_percent	Percent of text including spaCy PERCENT tags
ORDINAL_percent	Percent of text including spaCy ORDINAL tags
FAC_percent	Percent of text including spaCy FAC tags
LAW_percent	Percent of text including spaCy LAW tags

PRODUCT_percent	Percent of text including spaCy PRODUCT tags
EVENT_percent	Percent of text including spaCy EVENT tags
TIME_percent	Percent of text including spaCy TIME tags
LOC_percent	Percent of text including spaCy LOC tags
ORG_percent	Percent of text including spaCy ORG tags
WORK_OF_ART_percent	Percent of text including spaCy WOA tags
QUANTITY_percent	Percent of text including spaCy QUANTITY tags
LANGUAGE_percent	Percent of text including spaCy LANGUAGE tags
Max Word	Length of the longest word in the sentence
Min Word	Length of the shortest word in the sentence
Avg Word Length	Average length of words in the sentence
present_verb	Number of present tense verbs
past_verb	Number of past tense verbs
adjectives	Number of adjectives
pronouns	Number of pronouns
TO's	Number of to usages
determiners	Number of determiners
conjunctions	Number of conjunctions
dots	Number of (.) used
exclamations	Number of (!) used
question	Number of (?) used
ampersand	Number of (&) used

capitals	Number of capitalized letters
digits	Number of digits (0-9) used
long_word_freq	Number of long words
short_word_freq	Number of short words
followers_count	Number of followers
friends_count	Number of friends
favourites_count	Number of favourites across all tweets
statuses_count	Number of tweets
listed_count	Number of tweets the user has in lists
mentions	Number of times the user was mentioned
replies	Number of replies the user has
retweets	Number of retweets the user has
favourites	Number of favourites the user has
hashtags	Number of hashtags (#) the user has used
URLs	whether the user has a provided a url in relation to their profile
quotes	Number of times the user has been quote tweeted
BotScoreBinary	Binary score whether the user is considered a bot or not
cred	Credibility score
normalized_influence	Influence score the user has, normalized
majority_target	Truth value of the tweet
statement	Headline of a new article

BinaryNumTarget	Binary representation of the statement's truth value (1 = True / 0 = False)
tweet	Twitter posts related to the associated manual keywords

Truth_Seeker_Model_Dataset.csv

Name	Description
author	The author of the statement
statement	Headline of a new article
target	The groundtruth value of the statement
BinaryNumTarget	Binary representation of the target value (1 = True / 0 = False)
manual_keywords	Manually created keywords used to search twitter with
tweet	Twitter posts related to the associated manual keywords
5_label_majority_answer	Majority answer using 5 labels (Agree, Mostly Agree, Disagree, Mostly Disagree, Unrelated) *NO MAJORITY indicates that there was no consensus when a majority answer was generated.
3_label_majority_answer	Majority answer using 3 labels (Agree, Disagree, Unrelated) *NO MAJORITY indicates that there was no consensus when a majority answer was generated.

Contributing

The project is not currently in development, but any contribution is welcome. Please contact one of the authors of the paper.

Acknowledgments

The authors would like to thank the Canadian Institute for Cybersecurity (https://www.unb.ca/cic/about/index.html) for its financial and educational support.

Using the dataset

To learn more about why this dataset was created, watch this video, "Defending Democracy: Combatting Information Disorder by Sajjad Dadkhah (https://youtu.be/DVYrRQKZxsk)."

Citation

S. Dadkhah, X. Zhang, A. G. Weismann, A. Firouzi and A. A. Ghorbani, "The Largest Social Media Ground-Truth Dataset for Real/Fake Content: TruthSeeker (https://ieeexplore.ieee.org/abstract/document/10286332)," in *IEEE Transactions on Computational Social Systems*, 99. 1-15, Oct. 2023.

Download the dataset (http://cicresearch.ca/IOTDataset/TruthSeeker2023/)

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